

Yijiang Li

Assistant Computational Mathematician
Mathematics and Computer Science Division
Argonne National Laboratory
Lemont, IL 60439
(404)-660-1880
Email: yijiang.li@anl.gov

I. Earned Degrees

- Georgia Institute of Technology, Atlanta, GA, Ph.D. in Operations Research, 2023, advisor: Nikolaos V. Sahinidis and Santanu S. Dey
 - Dissertation: *Decomposition Algorithms for Certain Integer Problems over Networks*
- Georgia Institute of Technology, Atlanta, GA, B.S. in Industrial Engineering, 2018
- Georgia Institute of Technology, Atlanta, GA, B.S. in Aerospace Engineering, 2018

II. Employment History

- Argonne National Laboratory, July 2026–present
 - Assistant Computational Mathematician: Mathematics and Computer Science
- Argonne National Laboratory, August 2023–June 2026
 - Postdoctoral Appointee: Mathematics and Computer Science, advisor: Kibaek Kim
- Georgia Institute of Technology, August 2019–August 2023
 - Graduate Research Assistant: School of Industrial and Systems Engineering
- Amazon.com, May 2022–August 2022
 - Research Scientist Intern: Modeling and Optimization
- Argonne National Laboratory, May 2021–August 2021
 - Givens Associate: Mathematics and Computer Science
- Delta Air Lines, May 2016–August 2017
 - Co-op: Interiors Engineering

III. Publications

[†]Student or intern under my supervision.

A. Journal Articles

3. **Yijiang Li**, Santanu S. Dey, Nikolaos V. Sahinidis, Naresh Susarla, Miguel A. Zamarripa, and Markus A. Drouven. “Optimizing the Design and Operation of Water Networks: Two Decomposition Approaches”. In: *Computers & Chemical Engineering* 192 (2025), p. 108897. DOI: [10.1016/j.compchemeng.2024.108897](https://doi.org/10.1016/j.compchemeng.2024.108897).
2. **Yijiang Li**, Santanu S. Dey, and Nikolaos V. Sahinidis. “A Reformulation-Enumeration MINLP Algorithm for Gas Network Design”. In: *Journal of Global Optimization* 90.4 (2024), pp. 931–963. DOI: [10.1007/s10898-024-01424-x](https://doi.org/10.1007/s10898-024-01424-x).

1. **Yijiang Li**, John-Paul Clarke, and Santanu S. Dey. “Using Submodularity within Column Generation to Solve the Flight-to-Gate Assignment Problem”. In: *Transportation Research Part C: Emerging Technologies* 129 (2021), p. 103217. DOI: [10.1016/j.trc.2021.103217](https://doi.org/10.1016/j.trc.2021.103217).

B. Conference Proceedings

10. Olivera Kotevska, Max Carlson, Yan Gao, Francis Jeanson, **Yijiang Li**, William Linkskog, Mohammad Naseri, Minseok Ryu, Sahil Tyagi, Jerry Watkins, Feiyi Wang, Ravi Madduri, and Kibaek Kim. “From Pilots to Production: Lessons in Cross-Institutional Federated Training and Artificial Intelligence for Science”. In: *Proceedings of the SC’26 Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis (Trillion Parameter Workshops)*. To appear. 2026.
9. **Yijiang Li**, Emon Dey, Zilinghan Li, Krishnan Raghavan, Ravi K. Madduri, and Kibaek Kim. “FedQueue: Queue-Aware Federated Learning for Cross-Facility HPC Training”. In: *Proceedings of the 43rd International Conference on Machine Learning (ICML)*. 2026.
8. **Yijiang Li**, Emon Dey, Massimiliano Lupo Pasini, Teja Kuruganti, and Kibaek Kim. “Scaling Laws for Physics-Aware ACOPF Surrogate Learning”. In: *Proceedings of the SC’26 Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis (AI4S)*. To appear. 2026.
7. **Yijiang Li**, Zeeshan Memon, Hongwei Jin, Stefano Fenu, Keunju Song, Sunash B. Sharma, Parfait Gasana, Hongseok Kim, Liang Zhao, and Kibaek Kim. “LUMINA: Foundation Models for Topology Transferable ACOPF”. In: *ICLR 2026 Workshop on Foundation Models for Science (FM4Science)*. 2026.
6. Zeeshan Memon, **Yijiang Li**, Hongwei Jin, Kibaek Kim, and Liang Zhao. “Towards Systematic Generalization for Power Grid Optimization Problems”. In: *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), AI for Sciences Track. Oral presentation (top 6.1% of submissions)*. 2026.
5. Guangji Bai[†], **Yijiang Li**, Zilinghan Li, Liang Zhao, and Kibaek Kim. “FedSpaLLM: Federated Pruning of Large Language Models”. In: *Proceedings of the 2025 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*. 2025.
4. Zilinghan Li, Aditya Sinha, **Yijiang Li**, Kyle Chard, Kibaek Kim, and Ravi Madduri. “Experience Building Enterprise-level Privacy-Preserving Federated Learning to Power AI for Science”. In: *IEEE 7th International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (TPS-ISA)*. 2025.
3. Guangji Bai[†], **Yijiang Li**, Cheng Ling, Kibaek Kim, and Liang Zhao. “SparseLLM: Towards Global Pruning for Pretrained Language Models”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2024.
2. Shourya Bose, **Yijiang Li**, Amy Van Sant, Yu Zhang, and Kibaek Kim. “From RNNs to Foundation Models: An Empirical Study on Commercial Building Energy Consumption”. In: *NeurIPS 2024 Workshop on Time Series*. 2024.
1. Jian Li, Xiong Rao, Mao Tang, and **Yijiang Li**. “General Expression of Pressure Angle and Principal Curvature of Globoidal Indexing Cam Mechanism”. In: *IOP Conference Series: Materials Science and Engineering*. Vol. 474. 1. IOP Publishing. 2019, p. 012019. DOI: [10.1088/1757-899X/474/1/012019](https://doi.org/10.1088/1757-899X/474/1/012019).

C. Manuscripts

8. Stefano Fenu, Wei Gao, **Yijiang Li**, Hongwei Jin, Zeeshan Memon, Sunash B. Sharma, Liang Zhao, Jonghwan Kwon, and Kibaek Kim. *A Unified Multi-task Benchmark for Data-driven Models of Power Systems*. Under review. 2026.
7. Hongwei Jin, Stefano Fenu, **Yijiang Li**, and Kibaek Kim. *Harness for Multi-Facility HPC: Agent-Ready, not Agent-Resident*. Under review. 2026.
6. Hongwei Jin, Keunju Song, Zeeshan Memon, **Yijiang Li**, Stefano Fenu, Hongseok Kim, Liang Zhao, and Kibaek Kim. *LUMINA: A Grid Foundation Model for Benchmarking AC Optimal Power Flow Surrogate Learning*. 2026. arXiv: [2605.02133](#).
5. **Yijiang Li**, Zilinghan Li, Kyle Chard, Ian Foster, Todd Munson, Ravi Madduri, and Kibaek Kim. *Scalable Cross-Facility Federated Learning for Scientific Foundation Models on Multiple Supercomputers*. 2026. arXiv: [2603.19544](#).
4. Massimiliano Lupo Pasini, **Yijiang Li**, Kibaek Kim, and Teja Kuruganti. *Scalable Heterogeneous Graph Foundation Models for Data-Driven Optimal Power Flow in Smart Grids*. 2026. arXiv: [2605.23194](#).
3. Krishnan Raghavan, Emon Dey, Seok-Ju Hahn, **Yijiang Li**, Todd Munson, and Kibaek Kim. *Decoupling Federated Learning to Address Utility-driven Asynchrony*. Under review. 2026.
2. **Yijiang Li**, Kibaek Kim, Sven Leyffer, Matt Menickelly, Lawrence Paul. Lewis, and Joshua Bergerson. *Modeling and Solving Cascading Failures Across Interdependent Infrastructure Systems*. 2024. arXiv: [2407.16796](#).
1. **Yijiang Li**. *Decomposition Algorithms for Certain Integer Problems over Networks*. Ph.D. Thesis. 2023.

D. In Preparation

IV. Presentations

18. *Federated Learning Across HPC: What Deployment Taught Us*, FPAI-HPC'26 in conjunction with IEEE International Conference on Cluster Computing (CLUSTER'26), Alexandria, VA, September 2026.
17. *LUMINA: Graph Foundation Model for Power Grid Applications*, Trillion Parameter Consortium 2026 All-Hands Conference (TPC 26), Baltimore, MD, June 2026.
16. *Towards Scalable Federated Learning for Scientific Computing*, Trillion Parameter Consortium 2026 All-Hands Conference (TPC 26), Baltimore, MD, June 2026.
15. *Towards Scalable Federated Learning for Scientific Computing*, LANS Seminar, Argonne National Laboratory, Lemont, IL, March 2026.
14. *LUMINA: A Grid Foundation Model for Benchmarking AC Optimal Power Flow Surrogate Learning*, 5th GridFM Workshop, Boston, MA, March 2026.
13. *Scaling Federated Learning to HPC*, INFORMS Annual Meeting, Atlanta, GA, October 2025.

12. *Scalable Federated Learning Across DOE HPC Clusters*, Trillion Parameter Consortium 2025 All-Hands Conference (TPC 25), San Jose, CA, June 2025.
11. *Federated Learning Framework for Collaborative Training of Electric Grid Foundation Models*, Trillion Parameter Consortium 2025 All-Hands Conference (TPC 25), San Jose, CA, June 2025.
10. *FedSpaLLM: Federated Pruning of Large Language Models*, International Conference on Continuous Optimization (ICCOPT), Los Angeles, CA, June 2025.
9. *FedSpaLLM: Federated Pruning of Large Language Models*, SIAM Conference on Computational Science and Engineering, Fort Worth, TX, March 2025.
8. *FedSpaLLM: Federated Pruning of Large Language Models*, INFORMS Annual Meeting, Seattle, WA, October 2024.
7. *Gradient-Free Adaptive Global Pruning for Pre-trained Language Model*, Trillion Parameter Consortium, Model Architecture and Compute Performance Subgroup, Virtual, February 2024.
6. *Optimizing the Designs and Operations of Water Networks*, INFORMS Annual Meeting, Phoenix, AZ, October 2023.
5. *Optimization of Gas and Water Networks*, National Energy Technology Laboratory Stakeholder Meeting, Virtual, June 2023.
4. *Decomposition for Large-Scale Discrete Optimization*, Mathematics and Computer Science Division, Argonne National Laboratory, Lemont, IL, March 2023.
3. *A Decomposition Framework on Selecting Specialized Hubs*, INFORMS Annual Meeting, Indianapolis, IN, October 2022.
2. *Decomposition Framework for Solving Gas Network Expansion Optimization*, INFORMS Annual Meeting, Indianapolis, IN, October 2022.
1. *Using Submodularity within Column Generation to Solve the Flight-to-Gate Assignment Problem*, INFORMS Annual Meeting, Anaheim, CA, October 2021.

V. Grants

3. DOE ANL LDRD (\$50,000, 2026): Feasibility Study for Artificial Intelligence/Machine Learning-enabled Reactor Core Design and Analysis (Y. Li: co-PI). *[Unfunded]*
2. DOE ANL LDRD (\$50,000, 2025): Foundation Model for Coherent X-ray Imaging (Y. Li: co-PI). *[Successfully Completed]*
1. DOE ANL LDRD (\$50,000, 2024): Weather Informed Spatio-temporal Graph Neural Network for Power Grid (Y. Li: PI). *[Unfunded]*

VI. Other Activities

A. Teaching

1. Teaching Assistant for ISyE 4111 at Georgia Institute of Technology, Fall 2018.

B. Student Mentoring

Summer Students

1. Guangji Bai, Givens Associate, Emory University, 2024. Project: *Efficient Deployment of Large Language Models*. Joined Amazon, Seattle, Washington.

Research Mentoring

1. Sourabh Choudhary, Doctoral student, Georgia Institute of Technology, 2023.

C. Software

- **APPFL: Advanced Privacy-Preserving Federated Learning:** An open-source software framework for privacy-preserving federated learning that can be deployed for real FL experiments safely and easily among distributed environments. Yijiang Li is part of the core development team. (<https://github.com/APPFL/APPFL>)

D. Services

Membership

- Institute of Electrical and Electronics Engineers, 2026–present.
- Association for Computing Machinery, 2026–present.
- North American Chapter of the Association for Computational Linguistics, 2025–present.
- Society for Industrial and Applied Mathematics, 2024–present.
- Institute for Operations Research and the Management Sciences, 2018–present.
- American Institute of Aeronautics and Astronautics, 2018–2021.

Technical Journal or Conference Review

(number of papers refereed are shown in parentheses)

- *AAAI Conference on Artificial Intelligence* (4)
- *Transportation Research Part B: Methodological* (5)
- *Computers and Operations Research* (2)
- *International Conference on Machine Learning* (6)
- *Naval Research Logistics* (2)
- *Optimization and Engineering* (2)
- *IEEE Transactions on Power Systems* (2)
- *Computational Management Science* (2)
- *Conference on Neural Information Processing Systems* (4)
- *Pattern Recognition* (1)

Panels

- Panelist, *Federated and Privacy-Preserving AI for HPC*, FPAI-HPC'26 in conjunction with IEEE International Conference on Cluster Computing (CLUSTER'26), Alexandria, VA, September 2026.
- Panelist, *Preparing for Job Market (Academia, Industry, and National Labs)*, IISE Annual Conference and Expo 2026, Arlington, TX, May 2026.

VII. References

- **Dr. Kibaek Kim**, Computational Mathematician, Mathematics and Computer Science Division, Argonne National Laboratory, kimk@anl.gov.
- **Dr. Nikolaos V. Sahinidis**, Butler Family Chair and Professor, School of Industrial and Systems Engineering & School of Chemical and Biomolecular Engineering, Georgia Institute of Technology, nikos@gatech.edu.
- **Dr. Santanu S. Dey**, Anderson-Interface Professor and Director of Doctoral Recruiting and Admissions, School of Industrial and Systems Engineering, Georgia Institute of Technology, santanu.dey@isye.gatech.edu.